AERIS Cube — Project Overview

Oct 17, 2026 · @Nishchay Singh

Brief description

AERIS (Autonomous Experimental Reaction-wheel Inertial Stabilization) is a research platform for reaction-wheel attitude control. It centres on a cube designed to balance on an edge or a corner using three internal reaction wheels, the actuators satellites use to point themselves.

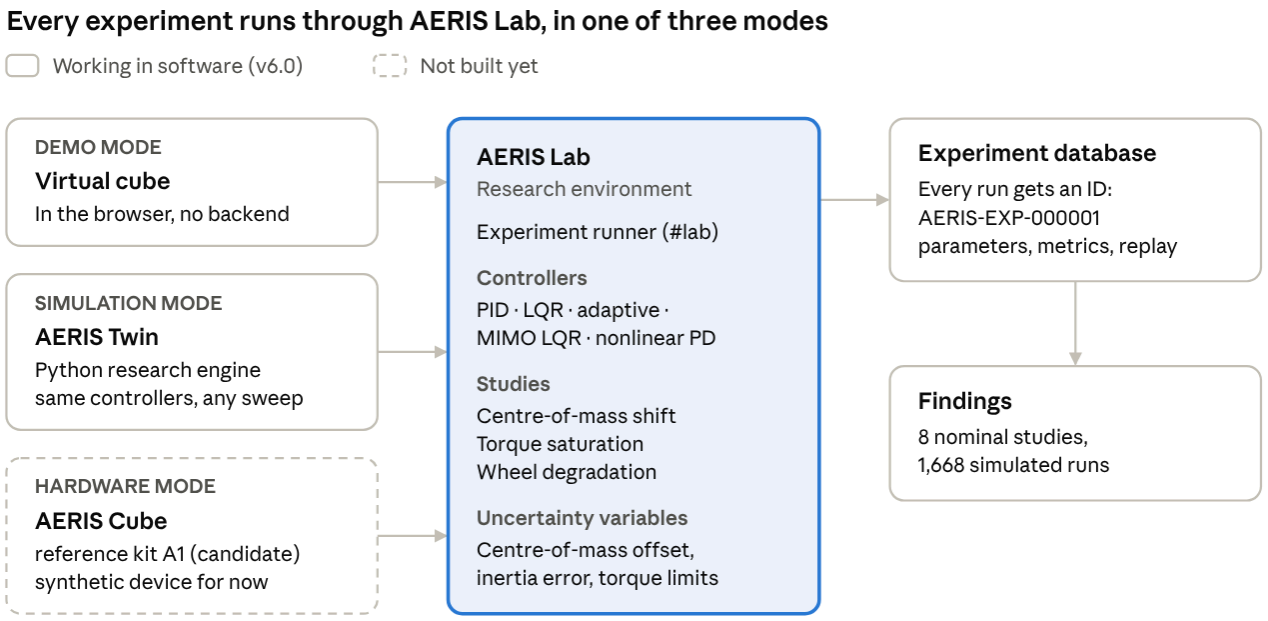
Today the software is real and tested: the AERIS Lab website with a virtual cube, a Python research backend, and C firmware for an ESP32-S3. The physical cube is a candidate design: none has been built, flashed or balanced yet (release v6.0, October 2026).

Its central question: how much uncertainty (centre-of-mass offset, inertia error, torque limits) can an attitude controller tolerate before stability is lost?

What it is

AERIS has three layers. AERIS Cube is the reference hardware design (kit A1), AERIS Twin is the simulation engine behind it, and AERIS Lab is the website where experiments run. The Lab works in three modes: DEMO (no backend), SIMULATION (the Python backend) and HARDWARE (live telemetry from a connected device).

Version 6.0 adds two journeys to the default virtual cube: Build Your Own Cube (the A1 kit, its bill of materials and sizing) and Connect My Cube (firmware install, pairing, calibration, telemetry and recordings). The project grew out of a single-axis build, the reaction wheel inverted pendulum (RWIP), and extends it from one axis to three.

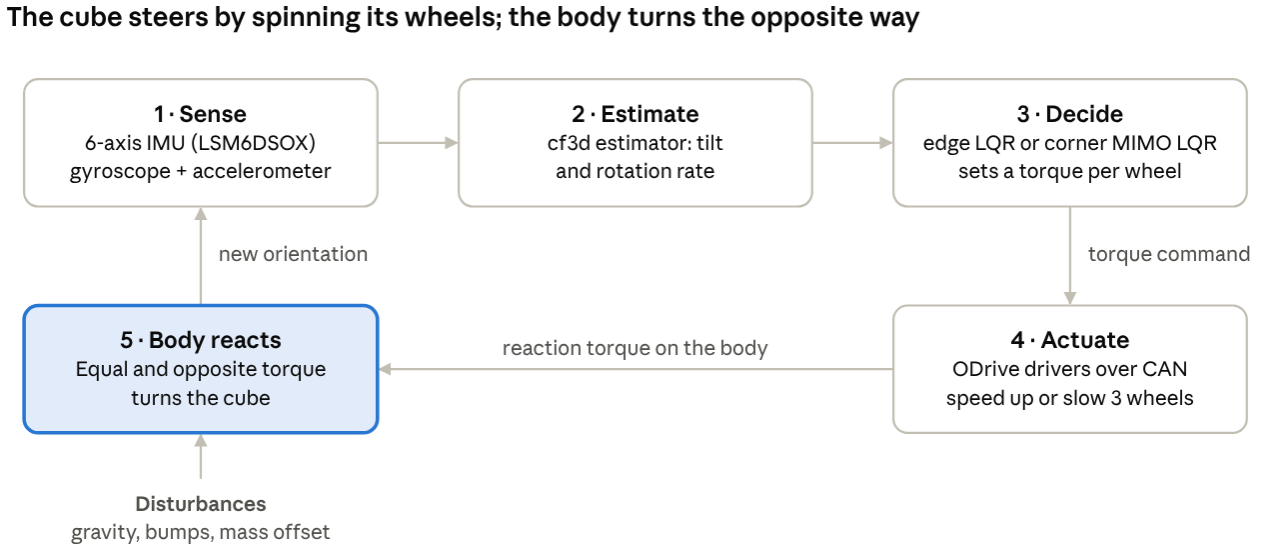


AERIS platform · three run modes into one lab

Every run goes through the Lab and lands in one experiment record (IDs like AERIS-EXP-000001). HARDWARE mode is exercised with synthetic devices for now, because no physical cube exists yet.

How it works

The cube turns itself by pushing against its own wheels: when a motor spins a wheel one way, the body turns the other way. With no outside torque, total angular momentum is conserved, so momentum moved into a wheel comes out of the body.



control loop · five steps, every cycle

Each cycle, 500 times a second in the firmware, the IMU reading becomes an estimate, the controller turns it into wheel torques, and the body's reaction closes the loop.

The physics

Ignoring friction and cross-axis coupling, the torque a motor applies to its wheel acts equally and oppositely on the body:

$$J\dot{\omega} = -\tau$$

Balancing on an edge adds gravity. Tilted by an angle θ from upright, the cube is an inverted pendulum, and wheel torque must counter gravity's tipping torque:

$$I\ddot{\theta} = m\,g\,\ell\sin\theta - \tau$$

J is the body's moment of inertia, ω its angular rate and τ the motor torque on the wheel. I is the inertia about the pivot edge, m the total mass and ℓ the pivot-to-centre-of-mass distance. On a corner the cube can tip about two axes at once, so all three wheels work together.

Sensing and estimation

A six-axis IMU (the LSM6DSOX is the current candidate) measures rotation rate (gyroscope) and acceleration, including gravity's direction (accelerometer). The firmware's cf3d estimator fuses them into one estimate of tilt and angular rate; the Lab also offers a multiplicative extended Kalman filter (MEKF). Gyroscopes are smooth but drift, and accelerometers are drift-free but noisy, so fusion keeps the strengths of each.

Control

500 times a second, the ESP32-S3 firmware turns the state estimate into a torque command for each wheel. On the device it runs an edge LQR and a corner MIMO LQR; the Lab compares five controllers on the same simulated plant:

- **PID** acts on the error, its running sum and its rate of change. Simple and model-free, but usually tuned axis by axis.
- **LQR** uses a linearised model to find a gain matrix K that balances error against effort, then applies u = −Kx to the whole state.
- **Adaptive** control updates itself as the plant changes, for example when the centre of mass shifts.
- **MIMO LQR** treats the three wheels as one coupled system, which is what corner balancing needs.
- **Nonlinear PD** acts on the full nonlinear attitude error rather than a linearised one.

Commands and telemetry travel over USB (Web Serial) or Wi-Fi (an outbound WebSocket over TLS), so each run can be logged and replayed in AERIS Lab.

The momentum limit

A reaction wheel can only spin so fast. A steady disturbance, such as an off-centre mass, keeps feeding momentum into the wheels until one saturates and control is lost. The firmware caps each wheel at its torque limit and gives no accelerating torque at its speed limit, and the Lab studies saturation directly.

Safety by design

- **Only a local press arms the motors.** Nothing sent from a computer or the website can arm them; a stop is accepted from any session, in every state.
- **Clear states.** BOOT → DISARMED → SELF_TEST → READY → ARMED → BALANCING → STOPPING. A fault is latched and clears only by holding ARM for 2 s once every cause is gone.
- **Protections while driving.** Stale or invalid IMU or driver data, over-current, over-speed, over-temperature, voltage limits, tilt beyond the abort angle and loop overruns all cut the drivers and latch a fault.
- **Release builds never drive motors.** Actuation exists only in experimental builds made from source, for a cube in a fall-catching fixture with a hardware stop chain.

Features

Everything below is implemented and tested in software (v6.0). The hardware entries describe the A1 reference design, which has not been built.

AERIS Cube — reference design A1 (candidate)

- **Three-axis actuation:** three reaction wheels, one per axis, each with its own motor driver (ODrive, over CAN).
- **Edge and corner balancing:** designed to hold an unstable equilibrium, not just an orientation.
- **Attitude sensing:** a six-axis IMU (LSM6DSOX candidate) on I2C.
- **On-board control:** an ESP32-S3 runs the control loop at 500 Hz.
- **Still to select:** motors, rotors, frame, harness, power system, local controls and the commissioning fixture.

Control

- **Firmware controllers:** a cf3d estimator, an edge LQR and a corner MIMO LQR, with gains computed by the backend and checked for parity with its Python.
- **Five controllers in the Lab:** PID, LQR, adaptive, MIMO LQR and nonlinear PD, compared on the same plant and seeds.
- **Saturation-aware allocation:** each wheel is capped at its torque limit, with no accelerating torque at its speed limit.

AERIS Twin — simulation engine

- **Research engine 1.1.0:** a rigid-body model of the cube, verified against an independent solver (RK4 order about 4, momentum conserved).
- **Studies:** 8 nominal studies and 1,668 simulated runs covering controller comparison, domains of attraction, centre-of-mass shift, saturation, wheel degradation, sensing noise and the MEKF estimator.
- **Simulate My Cube:** run the twin with a connected cube's own profile and compare the two.

AERIS Lab — research environment

- **Virtual cube by default:** the Lab (controllers, disturbances, sweeps, analysis) runs straight from a file, with no backend, account or hardware.
- **Three run modes:** DEMO, SIMULATION and HARDWARE.
- **Experiment records:** every run gets an ID such as AERIS-EXP-000001, with its parameters, metrics and replay.
- **Build Your Own Cube:** the A1 kit, a sourced bill of materials, the sizing calculation and guided steps.
- **Connect My Cube:** signed firmware install over Web Serial, pairing with a local press on the cube, diagnostics, calibration, bounded commands, telemetry and recordings.

Firmware — aeris-fw 0.1.0

- **Portable C11 core:** states, sensing, estimation, controllers, allocation, protection, the device link, commands and telemetry.
- **One core, three places:** ESP32-S3 images build with ESP-IDF v6.0.3; the test image passes 22 checks in the QEMU emulator; the same core runs in the browser as WebAssembly.
- **Tested:** 38 host tests, including every protection path, each proven by a control that removes the check and watches the test fail.

Uses and applications

AERIS is for anyone who needs to see how a reaction-wheel-controlled body behaves before trusting it with something expensive.

Use	What AERIS gives it
Satellite attitude control (ADCS) prototyping	Develop pointing algorithms on the virtual cube today, then on real hardware with real noise, friction and torque limits once the cube is built. CubeSats point with the same reaction-wheel principle.
Robust-control research	A repeatable way to measure how far centre-of-mass offset, inertia error or torque limits can go before a controller fails.
Controller benchmarking	Five controllers on the same plant and seeds, scored on the same criteria, every run stored and replayable.
Fault-tolerance studies	The wheel-degradation study shows whether control holds up when one wheel weakens.
Sim-to-real validation	Simulate My Cube runs the twin with a real cube's measured profile and compares the two.
Teaching	Angular momentum, rigid-body dynamics, state-space control and sensor fusion in a browser, with no install. Once built, the cube makes them visible on a desk.
Outreach and exhibitions	A cube balancing on its corner, once built, is an instant hook at science fairs and open days.

Potential

The cube is the entry point; the long-term value is a repeatable way to measure how and when attitude controllers fail.

- **A public benchmark.** The experiment records could become an open dataset of controller runs under controlled uncertainty, a reference others test against.
- **A remote control lab.** HARDWARE mode and Connect My Cube already handle pairing, leases and bounded commands, so one real cube could serve many students through a browser.

- **Learning-based control.** Logged runs and the twin give data and a safe sandbox to train or tune controllers (adaptive, reinforcement learning), then verify them on hardware.
- **Closer-to-orbit testing.** An air bearing would remove the pivot, friction and, with the mass balanced, gravity torque. The cube could then rehearse free three-axis manoeuvres the way a satellite flies them.
- **Momentum management.** The saturation study leads to the next problem real spacecraft solve: dumping stored wheel momentum, for example with magnetorquers.
- **A research result.** The studies already answer the core question in simulation: with an unknown centre-of-mass shift, PID recovers in 11% of runs and LQR in 75%. Hardware recordings can test whether that holds on a real cube.

Current status and specifications

The software is complete and tested at v6.0 (released October 2026); the cube itself is a candidate design that has not been built.

Part	Status	Detail
Website + AERIS Lab (virtual cube)	Working	DEMO and SIMULATION modes; browser test suites
Research backend + Twin	Working	Backend 1.2.0, research engine 1.1.0, 8 nominal studies
Connect My Cube + device API	Working	Tested with synthetic devices only; no physical board yet
Firmware aeris-fw 0.1.0	In development	38 host tests, WebAssembly, QEMU; never run on a physical board
Reference cube A1	Planned	Candidate: motors, rotors, frame, harness and power not selected
Hardware validation	Planned	Commissioning in a fall-catching fixture, measured profile, recorded trials

Hardware — A1 bill of materials (candidate)

Item	Qty	Status
Controller board (ESP32-S3)	1	candidate

Item	Qty	Status
IMU, six-axis (LSM6DSOX)	1	candidate
CAN transceiver and termination	1 + 2	candidate
Wheel motor driver (ODrive, CAN)	3	candidate
Wheel motor, rotor and encoder magnet	3 each	unresolved
Frame, guards, mounts and pivot pieces	1	unresolved
Wiring harness and power system	1 each	unresolved
Local controls (ARM, emergency stop, dead-man, LED)	1	unresolved
Commissioning fixture	1	unresolved

No kit price yet: it can't be stated before the motors, rotors, frame and power system are selected. Mass properties, rotor inertias and motor constants are still to be measured.

Software stack: a static website (HTML and JavaScript, runs from a file), a Python 3.12 backend (FastAPI, NumPy, SciPy, SQLite) and C11 firmware (ESP-IDF v6.0.3 for the ESP32-S3, clang for WebAssembly).

Simulated results (nominal parameters)

Study	Runs	Recovery rate
S1 controller comparison (corner)	150	PID 80% · LQR 80% · adaptive 80% · MIMO LQR 87% · nonlinear PD 80%
S3 unknown centre-of-mass shift (edge)	243	PID 11% · LQR 75% · adaptive 75%
S4a torque limits × initial tilt (edge)	135	PID 36% · LQR 33% · adaptive 33%
S4b wheel degradation (corner)	60	LQR 50% · MIMO LQR 50% · nonlinear PD 70%
S2 domain of attraction (edge)	663	39% for PID, LQR and adaptive alike

These describe the model with nominal parameters, not a physical cube; each rate covers only the sampled points of its study (studies/results/nominal-v5). The earlier benchmark on the project reference sheet (settling 6.21 / 3.02 / 1.85 s) has no traces and was not produced by this software, so it is not used.

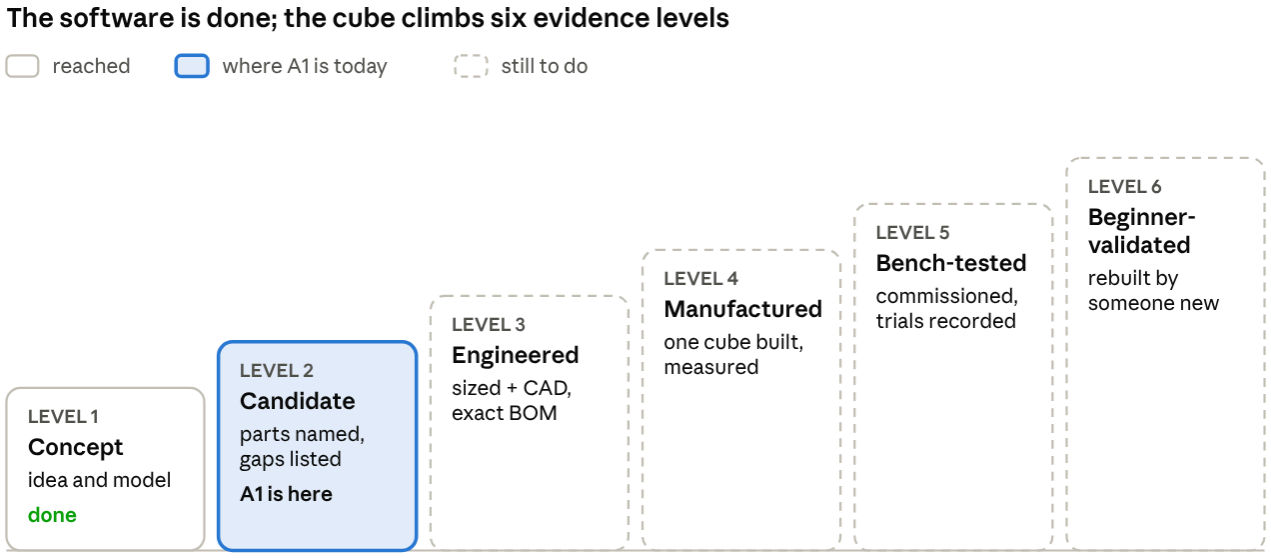
Limitations and roadmap

The main limits: the hardware doesn't exist yet, and every number so far comes from a model with nominal parameters.

- **No physical cube yet.** No A1 has been assembled, flashed, measured or balanced, and the firmware has never run on a physical board.
- **The model isn't validated.** Simulated results use nominal design parameters; a measured profile and real recordings are needed before they describe a real cube.
- **A desk is not orbit.** Balancing on an edge or corner is an inverted-pendulum problem under gravity; a satellite floats free. Results transfer as control methods, not as one-to-one flight predictions.
- **Wheels saturate.** Reaction wheels store momentum but cannot shed it, so a sustained disturbance eventually exhausts them.
- **Visuals are illustrations.** The renders and assembly video are AI-generated concept art, not photographs.

Roadmap

Hardware readiness is tracked as six evidence levels, and software cannot raise them. The A1 cube is at the second level, candidate.



A1 hardware readiness · 6 evidence levels

Motors may be energised only after commissioning: qualified limits, a fall-catching fixture and a wired hardware stop chain (hardware/COMMISSIONING.md).

Next steps

- [ ] Select the motor, rotor and driver together with the sizing tool (hardware/tools/sizing.py)
- [ ] Design the frame, guards, mounts and pivot pieces in CAD
- [ ] Design the power system, controller board and harness, and fix an exact BOM
- [ ] Build one A1 and measure its profile: mass properties, rotor inertias, motor constants
- [ ] Commission it in a fall-catching fixture with a hardware stop chain
- [ ] Record edge and corner trials and compare them with Simulate My Cube
- [ ] Film the real cube balancing, to replace the AI renders